



DEALER BEST PRACTICE SERIES

GAUGE TO ALIGN 3500  
ENGINE CAMSHAFT BEARINGS  
DURING INSTALLATION

Maintenance and Repair      Application      Component Life Management      Component Renewal (CRC)      MARC Management

**GAUGE TO ALIGN 3500 ENGINE CAMSHAFT BEARINGS  
DURING INSTALLATION.....0**

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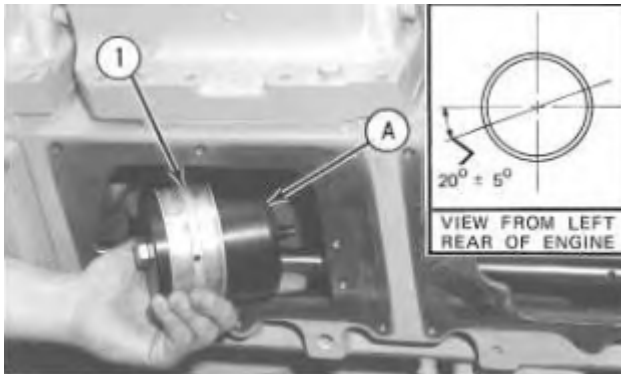
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## 1.0 Introduction

Camshaft bearings must be installed with the oil holes in the bearing aligned exactly with the oil passages in the cylinder in order to assure proper lubrication to the camshaft.

The install procedure states to position the oil hole on the bearing 20 degrees (plus or minus 5 degrees) off the horizontal centerline.

1. Install camshaft bearing (1) on tooling (A).



2. Position **camshaft bearing** (1) and Tooling (A) in the bore.

**Note:** The joint in **camshaft bearing** (1) must be on the horizontal centerline of the bore. The upper oil hole must be  $20 \pm 5$  degrees above the horizontal centerline of the bore. The lower oil hole of **camshaft bearing** (1) should be  $20 \pm 5$  degrees below the horizontal centerline of the bore. The joints of **camshaft bearing** (1) on the right side of the engine must face the inside of the cylinder block. The joints of **camshaft bearing** (1) on the left side of the engine must face the outside of the cylinder block.

This Best Practice provides a very simple gauge to assure proper alignment.

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Tool Guide for Installing Camshaft Bearings

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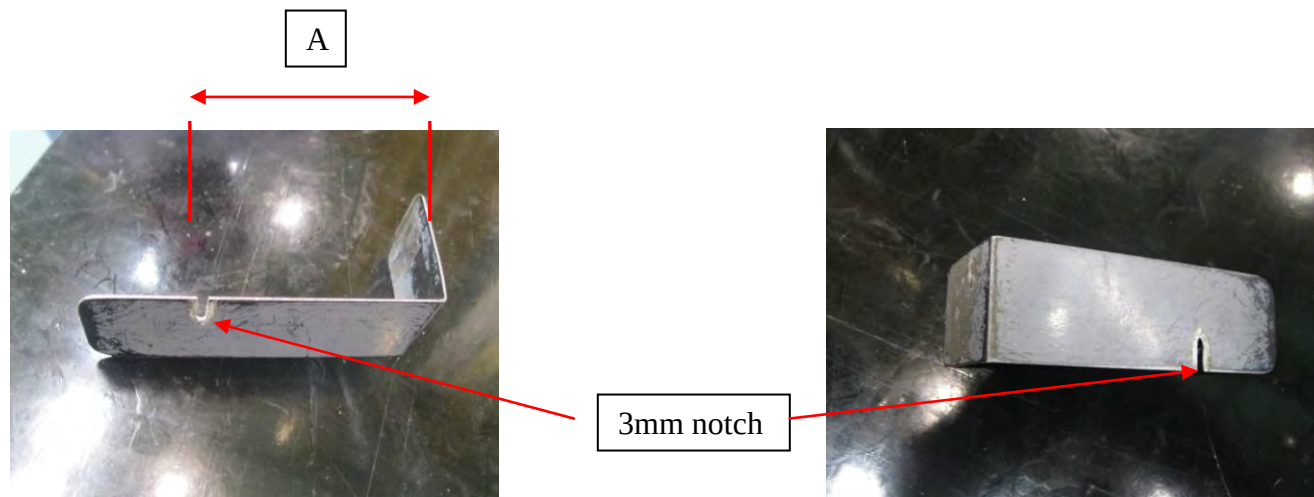
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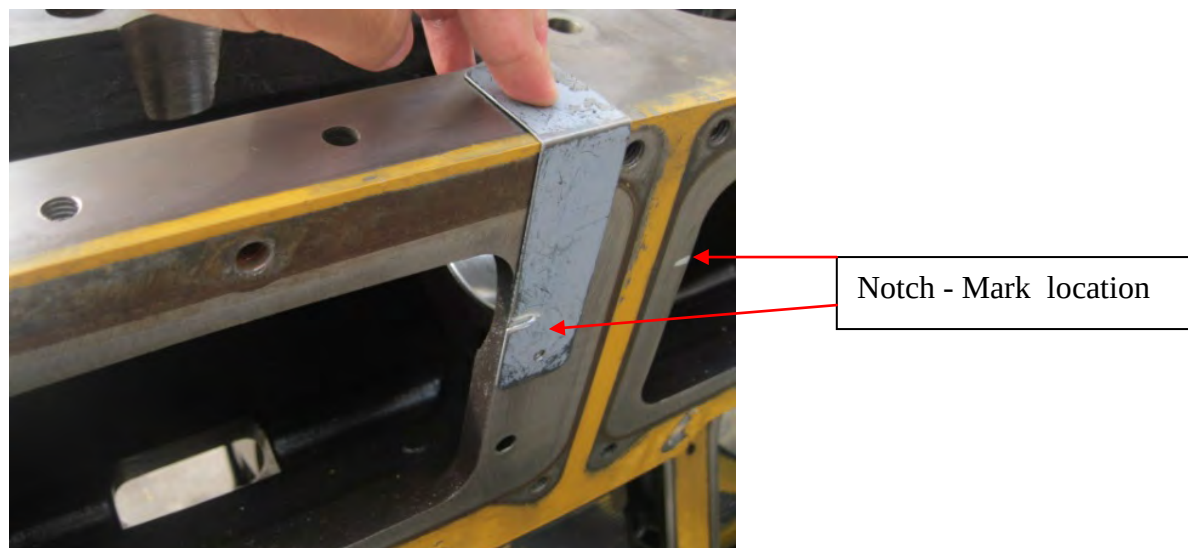
## 2.0 Best Practice Description

This is a very simple angle gauge to mark a position on the block to use to align the camshaft bearing oil hole.

Bend a steel plate at 90°. Make a notch of 3mm wide at a distance of  $92.00 \pm 0.06\text{mm}$  ( $3.620 \pm 0.002\text{inch}$ ) (A). The notch is used to locate the proper position on the block for alignment of the bearing oil hole. The block is marked with a fine point marking pencil.



Place the gauge on the edge of the block as shown and mark the bearing alignment location (notch) with a marking pencil



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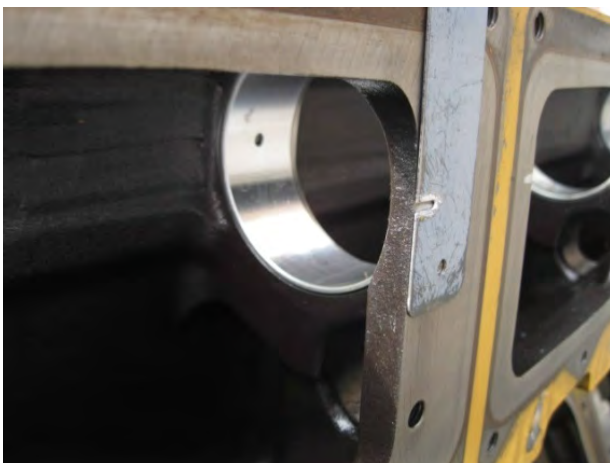
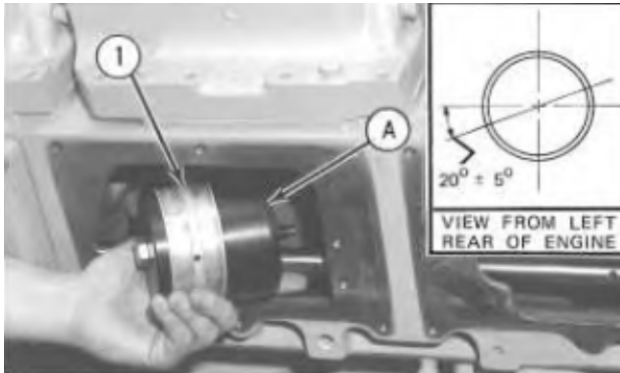
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Using the 8S2241 bearing installation tool, place the bearing on the installation tool. Rotate the installation tool and bearing to line up the bearing oil hole with the mark on the block. Install the bearing with press tool.



### 3.0 Implementation Steps

Fabricate the gauge as shown.

### 4.0 Benefits

- Quick, easy and correct installation of camshaft bearings.

### 5.0 Resources Required

Minimal expense to fabricate the tool.

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## 6.0 Supporting Attachments / References

None.

## 7.0 Related Best Practices

None.

## 8.0 Acknowledgements

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